

Sustainable cationization of bark tannins from native Brazilian tree species using hexamine as a formaldehyde substitute for water treatment applications

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ABSTRACT

The growing concern over the environmental and health risks of conventional coagulants has driven interest in bio-based alternatives. This study evaluated the potential of tannins extracted from the bark of *Stryphnodendron adstringens* and *Mimosa tenuiflora*, chemically modified by cationization, for application as natural coagulants in water treatment. Chemical characterization included total extractives, phenolic content, flavonoids, condensed tannins, and the Stiasny index. Cationization was performed using the Mannich reaction (with formaldehyde) and an alternative method with hexamethylenetetramine (hexamine). FTIR was used to confirm structural changes, and coagulation performance was assessed in water with 150 NTU turbidity. *M. tenuiflora* showed higher phenolic content (135.74 mg GAE/g), while *S. adstringens* had a higher condensed tannin content (15.23%) and Stiasny index (93.18%), indicating greater tannin purity. Tannins from *M. tenuiflora* cationized via the Mannich reaction removed over 99% turbidity, outperforming ferric chloride and matching commercial tannins. Hexamine-based cationization yielded removals of 59% (*M. tenuiflora*) and 64% (*S. adstringens*). FTIR confirmed successful incorporation of amine groups and structural differences between the modification methods. These results demonstrate the potential of native tannins as effective and sustainable natural coagulants. Although hexamine-based modification showed slightly lower performance than formaldehyde, it presents a viable, less toxic alternative for future development.

1. Introduction

The sustainable management of water resources has increasingly become a central issue in recent years, particularly concerning their availability, reuse, and the environmental impact of improper effluent disposal. In this context, there is a growing demand for innovative technologies that enable efficient and environmentally sustainable water treatment [1]. Among the fundamental stages of water treatment, the coagulation/flocculation process stands out. This process involves the destabilization of colloidal particles through the addition of a coagulant with an opposite charge under rapid mixing. The coagulant

facilitates the aggregation of these particles, leading to the formation of larger and denser flocs (flocculation), which, under slow mixing, become sedimentable and can subsequently be removed through the sedimentation process [2,3].

The predominant coagulants utilized in the industry are those derived from chemical processes, including aluminum sulfate and ferric chloride. Nonetheless, their use is accompanied by several disadvantages, such as the risk of soil toxicity [4], a notable decrease in water pH that can result in corrosion and scaling of treatment equipment [5], and potential health concerns, particularly the link between aluminum exposure and Alzheimer's disease [6]. Considering these issues, there is

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an increasing interest in investigating alternative coagulants sourced from natural and renewable materials, such as tannin-based coagulants, which may serve as substitutes for chemical coagulants in certain applications while providing both environmental and operational advantages [2].

Tannins are polyphenolic compounds belonging to a broad class of secondary metabolites, typically stored in plant cell vacuoles in the form of esters or glycosides [7]. These compounds can be classified into two main groups: condensed tannins and hydrolyzable tannins. The former are primarily composed of proanthocyanidins, while the latter consist of polyesters derived from gallic acid or hexahydroxydiphenic acid and their analogs [8,9]. Condensed tannins are widely found in plant bark [10], whereas hydrolyzable tannins occur in various plant parts, including wood, bark, leaves, fruits, and branches [9]. Additionally, these compounds exhibit high chelating capacity due to the presence of multiple hydroxyl groups and can undergo various chemical modifications, enhancing their potential application in the coagulation of colloidal particles [11,12].

Due to the anionic nature of tannins and the predominance of negatively charged colloidal particles in natural and wastewater, the functionalization of these compounds for application as coagulants generally requires a chemical modification process known as cationization [3], which involves the Mannich-based reaction.

The cationization of tannins is often carried out through the Mannich reaction, a widely used method for introducing amine groups into their structure. This process can involve compounds such as ethanolamine, diethanolamine, methylamine, or ammonium chloride, along with an aldehyde, with formaldehyde being the most used [13]. In this reaction, the interaction between formaldehyde and the amine generates an iminium ion, which subsequently reacts with the phenolic ring of the tannin, promoting the substitution of a hydrogen atom [2,11]. As a result, the weight of the polymer increases, and positive charges are introduced into its structure, enabling the cationic coagulants to destabilize and neutralize anionic colloidal particles present in water [3,13].

However, both formaldehyde and the amines used in the cationization process are considered toxic compounds [14]. Additionally, the reaction may result in the presence of unreacted residues in the final product, posing environmental risks and potential health impacts [15]. According to the International Agency for Research on Cancer (IARC), formaldehyde is classified as a leukemogenic agent in humans, with exposure being linked to the development of various types of cancer [16]. Given these concerns, it is essential to seek alternatives to formaldehyde for its replacement in the Mannich reaction, aiming to reduce the negative impacts associated with its use.

Dhawale et al. [17] demonstrated that the use of hexamethylenetetramine (hexamine) significantly reduces formaldehyde emissions compared to its direct application. According to the authors, this effect occurs because hexamine releases formaldehyde in a controlled and gradual manner, minimizing its presence in the final product. Additionally, hexamine exhibits high reactivity with tannins, interacting directly with their phenolic groups [18]. Thus, its use emerges as a promising alternative to replace formaldehyde, offering a potentially less toxic and more environmentally sustainable process. In this context, investigating the feasibility of tannin cationization using hexamine is highly relevant, especially when considering tannin-rich species such as *Stryphnodendron adstringens* (Mart.) Coville and *Mimosa tenuiflora* (Willd.) Poir., native to the Cerrado and Caatinga biomes, respectively.

In the search for formaldehyde substitutes, Machado et al. [1] observed promising results in the removal of turbidity and color from effluents by using ammonium hydroxide as an alternative to formaldehyde in their reaction. Similarly, Bello and Leiviskä [14] investigated the use of melamine as a nitrogenating source in the cationization of tannins, as well as the effects of variations in the melamine-to-formaldehyde ratio and changes in activation conditions. Recent studies have already pointed out the risks associated with the extensive use of conventional and non-renewable coagulants in water treatment,

which may impair the final water quality and pose risks to human health [13]. However, most studies focus only on reducing formaldehyde usage rather than on its complete replacement in the process. The main challenge is to provide a material with performance comparable to that of conventional products while employing more sustainable techniques. By doing so, it will be possible to maintain effective coagulation performance and promote greater sustainability in the water treatment sector.

These studies highlight the potential of different compounds to act synergistically with tannins to enhance coagulation efficiency, emphasizing the importance of further research to identify equally viable and environmentally friendly alternatives. Therefore, the use of hexamine in this study aligns with green chemistry principles by minimizing the ecological footprint of the coagulant and reducing risks related to residual toxicity. Our study stands out by bringing to the discussion more sustainable alternatives for water treatment, aiming to move toward solutions that are more strongly focused on natural products and their potential. In doing so, we seek to contribute to the achievement of the Sustainable Development Goals, particularly Goal 6 (Clean Water and Sanitation) and Goal 13 (Climate Action).

The aim of this study is to chemically modify the tannins from *Stryphnodendron adstringens* and *Mimosa tenuiflora*, with and without the use of formaldehyde, and to perform the chemical characterization of the tannins from these species. Additionally, the chemical structure of the tannins will be characterized after cationization, considering the modifications introduced. The performance of the modified tannins will be tested for turbidity removal in water supplies. The goal is to identify a chemical modification that is as efficient as those currently used but less harmful to the environment, contributing to the development of sustainable and effective coagulants for water treatment.

2. Material and methods

2.1. Sampling and preparation of material

The bark was collected from individuals of *Stryphnodendron adstringens*, located in areas of forest fragments, with an average diameter of 15.28 cm, in the municipality of Lavras, Minas Gerais, Brazil (21°14'43" S, 44°59'59" W), with a climate of type Cwa according to Köppen. The bark of the *Mimosa tenuiflora* individuals was obtained from a 6-year-old commercial plantation with an average diameter of 13.25 cm, collected in the municipality of Macaíba, Rio Grande do Norte, Brazil (5°51'30" S and 35°21'14" W), with a climate in transition between types As and BSw, according to the Köppen classification. The bark of both species was removed lengthways along the trunk using machetes and packed in plastic bags. They were then dried at room temperature for two weeks and then ground in a Retsch SK hammer mill to obtain a finer, more uniform material. The ground bark was sorted using sieves to extract the material.

2.2. Extraction of tannins

The extractions were carried out at the Wood Anatomy and Chemical Products Laboratory at the Federal University of Lavras and were done in two ways: one to determine the Stiasny Index and condensed tannin content, and the other to obtain powdered tannins for the subsequent cationization process. Extractions to determine the Stiasny index and condensed tannin content were based on the methodology carried out by Mori et al. [19], which used 100 g of dry bark, 1500 mL of water (liquor/bark ratio 15:1, v/p) and 3% sodium sulphite (Na₂SO₃) in a water bath at 70 °C for 3 h. Extraction to obtain the powdered tannins was carried out in a similar way. However, after filtering using a vacuum pump, the liquid was taken to an oven at 45 ± 3 °C until the dry material was obtained. Finally, this material was macerated using a mortar and pestle to a particle size of 200 mesh.

2.3. Chemical characterization of the bark extracts

The Stiasny index was determined following the method of Guangcheng et al. [20]. A 20 g sample of concentrated extract was mixed with distilled water, formaldehyde (37%, m/m), and 10 N HCl, then heated under reflux for 35 min. This process formed insoluble tannin complexes, which were separated by filtration and dried at $103 \pm 2^\circ\text{C}$ until constant weight. The total solids content was measured by drying a separate 20 g extract sample at $105 \pm 3^\circ\text{C}$ for 24 h [21]. The Stiasny index (I) was calculated by dividing the dry mass of the precipitate by the total solid mass, and the total solids content (TST) and condensed tannin content (TTC) were determined based on the Stiasny index and total solids yield.

The total phenolic content was measured using the Folin-Ciocalteu method [22], where a 100 μL extract sample was reacted with the reagent and sodium carbonate solution, and absorbance at 760 nm was used to determine the phenolic content in gallic acid equivalents (GAE). The flavonoid content was determined by a colorimetric assay using aluminum chloride [23], with results expressed in catechin equivalents (CE). The total extractive content was quantified following TAPPI T204 cm-97 [24].

2.4. Chemical modification of tannins (Cationization)

To improve the coagulation properties of tannins, a chemical modification process aimed at their cationization is recommended. This process was carried out in two ways for the tannins from the two species under study, with the first method based on the Mannich reaction, as reported by Konrath and Fava [25], and consisting of three steps. In the first step, 5.4 g of ammonium chloride and 24.4 g of formaldehyde were placed in a volumetric flask, and the mixture was heated at 80°C for 2 h. The effectiveness of the reaction between formaldehyde and ammonium chloride was visually verified by a color change from light yellow to colorless. In the second step, the product obtained was mixed with 28.0 g of a 50% aqueous tannin solution and heated for 30 min at 60°C . The final stage consisted of adding 0.2 g of monoethanolamine, which reacted for 3 h at 50°C . The second method of cationization was carried out without the use of formaldehyde, also based on the Mannich reaction but with modifications. In this case, formaldehyde was replaced by hexamethylenetetramine (hexamine), used in a 50% aqueous solution to cationize the tannins.

2.5. Fourier Transform Infrared Spectroscopy (FTIR) Analysis

The FTIR technique was applied to characterize the tannins before and after the cationization process, with the aim of identifying changes in the functional groups and molecular structures. The analysis allowed for the observation of chemical modifications resulting from the process. The analysis was conducted at the Chemistry Laboratory of the Federal University of Lavras, using a Shimadzu IRAffinity⁻¹ spectrophotometer. The samples were dried in an oven at 60°C for three hours, and the spectra were obtained by attenuated total reflectance (ATR). Each spectrum was generated with 32 scans, with a resolution of 4 cm^{-1} , in the range of 4000 to 500 cm^{-1} [26].

2.6. Method for evaluating the coagulating potential of tannins for water treatment

Water samples were collected from the reservoir at the Escola Agrícola de Jundiá, in the municipality of Macaíba/RN, and the tests were conducted at the Environmental Microbiology Laboratory, within the Technological Complex of Engineering at the Federal University of Rio Grande do Norte (UFRN). For the analysis of the coagulation/flocculation process, the Jar-test equipment was used. In each jar, 1.5 L of the collected water was added, and the initial pH and turbidity were measured. The coagulants used were cationized tannins, with and

without formaldehyde, from *Stryphnodendron adstringens* and *Mimosa tenuiflora*, the commercial tannin from *Acacia mearnsii* (black wattle commercial tannins), and, for comparison, ferric chloride was also evaluated. The following combinations of coagulant concentrations and agitation speeds (coagulation and flocculation times) were applied for the two species, selected based on preliminary tests: 50 mg at 100 rpm for 3 min and 30 rpm for 30 min, and 150 mg at 100 rpm for 1 min and 30 rpm for 10 min. One hour after the completion of each test, the final pH and turbidity of each sample were assessed to determine which combination of variables resulted in the best turbidity removal.

2.7. Statistical analysis

The statistical analysis was performed with the data tabulated and processed using Origin® 2018 software (Massachusetts, USA). The FTIR analysis and coagulation test graphs were generated from this process. After the creation of the images, they were edited and enhanced using CorelDRAW 2020 software to ensure a clearer and more appropriate visual presentation of the results.

3. Results and discussion

3.1. Chemical characterization of the bark extracts

The bark extracts of *Stryphnodendron adstringens* and *Mimosa tenuiflora* exhibited differences in their chemical composition, as shown in Table 1. The total extractives content was higher in *M. tenuiflora* (26.94%) compared to *S. adstringens* (23.27%), which was also reflected in the higher concentration of total phenols (135.74 mg GAE/g of extract). Flavonoid analysis indicated similar values for both species, with a slightly higher concentration in *M. tenuiflora*. Additionally, *S. adstringens* presented a higher Stiasny Index (93.18%), suggesting a greater proportion of condensed tannins relative to other extract components. However, the absolute content of condensed tannins was similar between the species, with values of 15.23% for *S. adstringens* and 13.56% for *M. tenuiflora*. The total solids content was also higher in *M. tenuiflora*, while the non-tannin fraction was more pronounced in this species, suggesting the presence of other non-phenolic compounds that may influence its coagulating properties.

The extracts obtained from the bark of *Stryphnodendron adstringens* and *Mimosa tenuiflora* showed a high concentration of total extractives. This result is expected, as the bark is typically the plant tissue with the highest levels of secondary metabolites, functioning as a natural barrier against pests, pathogens, and environmental stressors [8,27,28]. Among the extractives are important classes of metabolites, such as terpenes, alkaloids, flavonoids, and tannins [29], which confer relevant chemical and bioactive properties to the bark.

Regarding tannins, the content of condensed tannins exceeded 10% in both species, indicating potential for commercial exploitation, as this threshold is considered adequate for large-scale extraction [30]. The Stiasny index was significantly higher in *S. adstringens*, reflecting greater

Table 1

Chemical composition of *Mimosa tenuiflora* and *Stryphnodendron adstringens* bark extract.

Analysis	Extracts	
	<i>Stryphnodendron adstringens</i>	<i>Mimosa tenuiflora</i>
Total extractives (%)	23.27 $\pm$ 1.56	26.94 $\pm$ 2.02
Total phenols (mg GAE/g of extract)	93.41 $\pm$ 5.31	135.74 $\pm$ 10.49
Flavonoid (mg CE/g of extract)	130.22 $\pm$ 8.25	138.57 $\pm$ 5.87
Total solids (%)	10.78 $\pm$ 1.01	13.97 $\pm$ 0.98
Stiasny Index (%)	93.18 $\pm$ 2.92	64 $\pm$ 3.11
Condensed tannins (%)	15.23 $\pm$ 0.89	13.56 $\pm$ 1.05
Non-tannin (%)	1.12 $\pm$ 0.19	7.63 $\pm$ 0.77

extract purity, i.e., a higher proportion of tannins relative to other compounds. The total phenolic contents were similar to those reported for the bark of *Anacardium occidentale* (92.2 mg GAE/g of extract) [26], but considerably lower than those found in *Acacia mearnsii* (589 mg GAE/g) [31], one of the main commercial sources of tannins. As a subset of total phenolics, tannins stand out due to their ability to form complexes with colloidal particles, making them promising candidates for use as coagulants in water treatment [32]. Therefore, based on the chemical characterization of the bark, the studied species show potential for application in environmental technologies, such as water and effluent treatment, as investigated in the present study.

3.2. Chemical structure of tannins with and without chemical modification

The functional groups of pure *S. adstringens* tannins, extracted in hot water, modified by the Mannich reaction and by hexamine were analyzed by FTIR. The spectrum is shown in Fig. 1 and their assignments in Table 2. This method is not able to fully elucidate the chemical composition of tannins, but it is possible to get an idea of the main functional groups present [33,34].

In this study, the band referring to O—H stretching in phenolic and aliphatic structures was identified at 3199 cm^{-1} for pure *S. adstringens* and at 3361 cm^{-1} and 3351 cm^{-1} for the species cationized through the Mannich and hexamine reaction, respectively, identifying that the latter two can be attributed to the elongation of primary amides, indicating that N—H bonds have been formed [35] and that the amine group was successfully added to the tannin [36], as well as having more hydroxyl groups in their molecules due to the use of formaldehyde and hexamine in the reaction [37].

The peak at 1608 , 1612 and 1616 cm^{-1} , for pure *S. adstringens*, cationized by Mannich reaction and hexamine, is a characteristic band of the C=C stretching vibration in the aromatic ring, in which the last two showed greater band intensity, due to the ring being more exposed when the structure is opened for modification [38,39]. According to Ntenga et al. [40], the high C=C peaks, combined with the species' high Stiasny index, can be considered an indicator of the purity of tannin extracts.

The peak at 1527 cm^{-1} found for pure *S. adstringens* disappears in the modified tannins. This peak shows the presence of condensed tannins in

Table 2

FTIR summary of pure *S. adstringens* tannins, chemically modified using the Mannich reaction and modified with hexamine.

Peaks (cm^{-1})			
<i>S. adstringens</i>	<i>S. adstringens</i> Mannich reaction	<i>S. adstringens</i> Hexamine	Assignment
—	3361	3351	OH stretching in hydrogen bonds
3199	—	—	OH stretching in hydrogen bonds
2346	2350	2346	Elongation N=C=O
1608	1612	1616	Aromatic C=C bonds
1527	—	—	Aromatic C=C bonds
1440	1450	1444	Vibrational movements of the C=C groups in the aromatic rings, aromatic C—H bending, C—O stretching and C—OH deformation
1334	1334	—	Saturated C—H in plane bending vibration
—	1218	1245	Lengthening of the C—O carboxyl group, which means gallic acid or some carbohydrate residues
—	1108	1112	Aromatic C—H in plane bending vibration and the C—O stretching for the OH substitution of epicatechin in the pyran ring
1031	—	1012	Stretching vibrations C—O or epicatechin and proanthocyanidins

the molecule [40]. The peaks located at 1440 cm^{-1} (pure *S. adstringens*), 1450 cm^{-1} (Mannich reaction) and 1444 cm^{-1} (hexamine) result from the combination of aromatic C—H bending, C—O stretching and C—OH deformation [41]. The band previously at 1440 cm^{-1} was shifted to 1450 and 1444 cm^{-1} after their respective modifications, indicating the overlap with the absorption coming from the asymmetric bending of the CH groups [38].

The peaks at 1334 cm^{-1} were observed for pure and cationized *S. adstringens* via the Mannich reaction, disappearing when using hexamine, and are attributed to saturated C—H in plane bending vibration [42]. Two new bands were observed only after the modification of the tannins, at 1218 and 1245 cm^{-1} and 1108 and 1112 cm^{-1} , for the Mannich

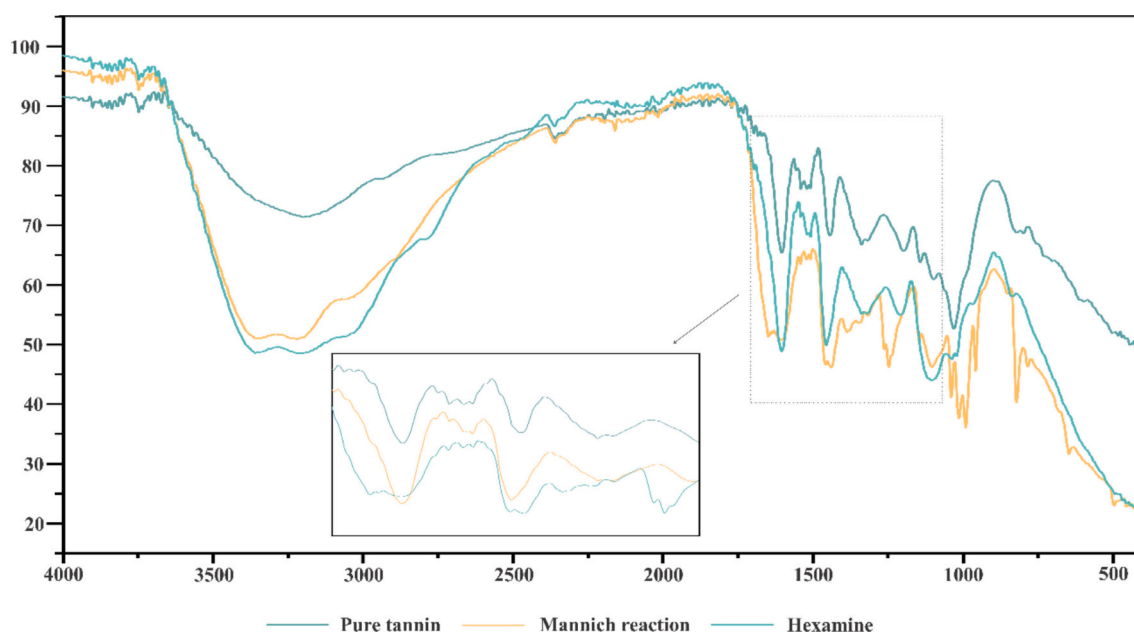


Fig. 1. FTIR spectra of tannins from *S. adstringens*.

reaction and hexamine, respectively, in which, in condensed tannins, the signals in the region between 1300 and 1200 cm^{-1} show the vibrations of the B ring, which can be attributed to the catechol-like vibrations of the B ring, while the region between 1200 and 1100 cm^{-1} shows the vibrations of the A ring, similar to the resorcinol vibration of the A ring [39].

The functional groups of pure *M. tenuiflora* tannins, extracted in hot water, modified by the Mannich reaction and by hexamine were analyzed by FTIR. The spectrum is shown in Fig. 2 and their assignments in Table 3. As with *S. adstringens*, it is possible to confirm that the amine group was added to the tannins, since the peaks were only added to the reaction after cationization, where for the Mannich reaction the peak was at 3154 cm^{-1} and for hexamine it was at 3203 cm^{-1} , in which the latter showed a more intense stretching. For these peaks, there is a typical broad band for O—H stretching, providing information about the occurrence of the polymerization process.

In the pure tannins of *M. tenuiflora*, a small vibrational band was observed at 2985 cm^{-1} , which refers to the vibrational movements of C—H stretching, was present in the pure tannins of *M. tenuiflora* and disappeared after the cationizations carried out. Because the band represents a smaller peak, it is possible that the interactions and degree of polymerization did not significantly affect the vibrational frequencies of the molecular structures and the position and intensity of the band was not highly modified [36,43].

Following the same pattern as *S. adstringens*, it was possible to observe that the peaks 1241 and 1243 cm^{-1} appeared after the cationizations carried out, both for the Mannich reaction and the hexamine reaction. For the hexamine cationization, two peaks appeared, at 1004 and 802 cm^{-1} , where the former is attributed to an intensity in the C—N bonds, probably due to the ammonium chloride in the cationization reaction [44], and the latter is attributed to the OH movements of aromatic alcohols and out-of-plane bending of aromatic CH [45].

3.3. Evaluating the coagulating potential of tannins for water treatment

The initial measurements of pH and turbidity in the water were recorded prior to the trials involving coagulation agents, yielding values of 6.54 and 150 NTU, respectively. The water temperature was in the

Table 3

FTIR summary of pure *M. tenuiflora* tannins, chemically modified using the Mannich reaction and modified with hexamine.

Peaks (cm^{-1})			
<i>M. tenuiflora</i>	<i>M. tenuiflora</i> Mannich reaction	<i>M. tenuiflora</i> Hexamine	Assignment
—	3154	3203	OH stretching in hydrogen bonds
2985	—	—	C-H stretching vibration in aromatic methoxyl groups and in methyl and methylene groups of side chains
2343	2354	2344	Elongation N=C=O
1608	1610	1617	C-H bond
1529	—	—	Aromatic C=C bonds
1444	1444	1444	Vibrational movements of C = C groups in aromatic rings, aromatic C—H bending, C—O stretching and C-OH deformation
1334	1322	—	Saturated C—H in plane bending vibration
—	1241	1243	Vibrational movements of C=C groups in aromatic rings
1101	1099	—	Aromatic C—H in plane bending vibration and the C - O stretching for the OH substitution of epicatechin in the pyran ring
—	—	1004	Vibrational movements of C=C groups in aromatic rings
—	—	802	Deformation movements of the aromatic C- H ring

range of 25 °C (ambient temperature). As illustrated in Fig. 3, the final pH of the water was assessed following experiments with various coagulants at a concentration of 50 mg, which underwent rapid mixing at 100 rpm for 3 min, succeeded by slow mixing at 30 rpm for 30 min. Fig. 3 also shows the percentage of turbidity reduction achieved under the same experimental parameters. It was noted that only the inorganic coagulant, ferric chloride, led to a significant decrease in the water's pH,

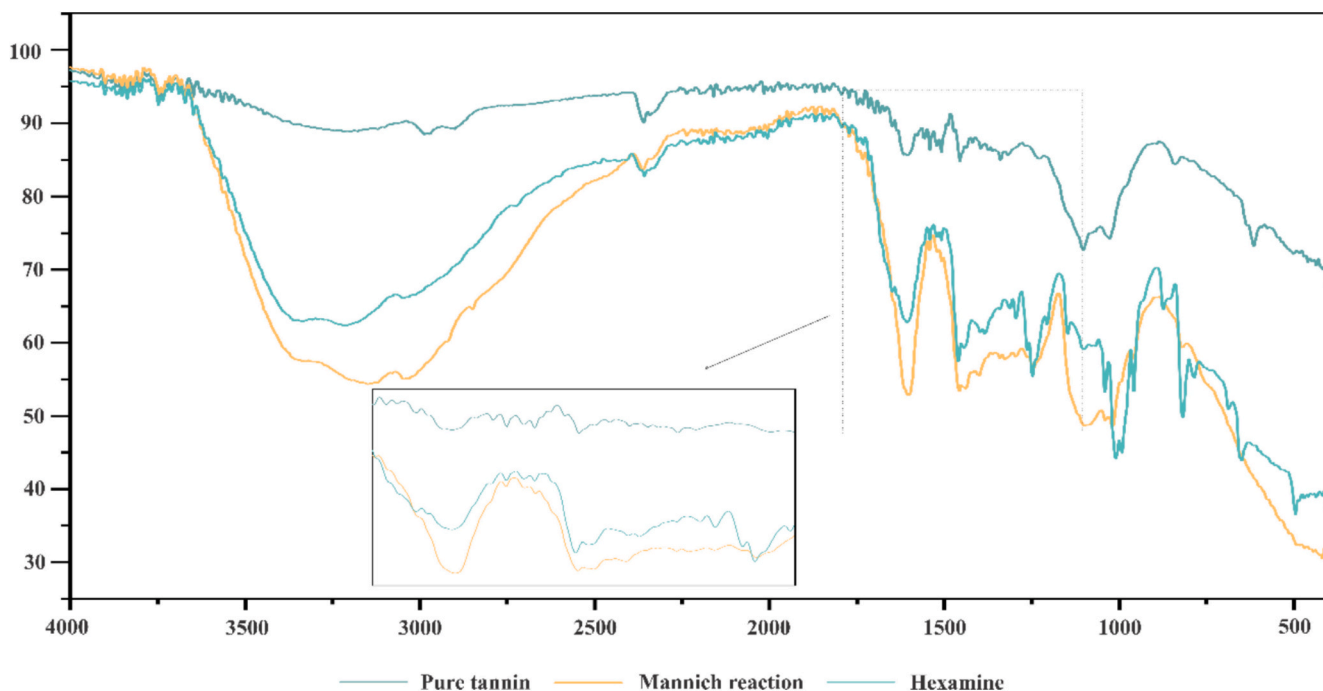


Fig. 2. FTIR spectra of tannins from *M. tenuiflora*.

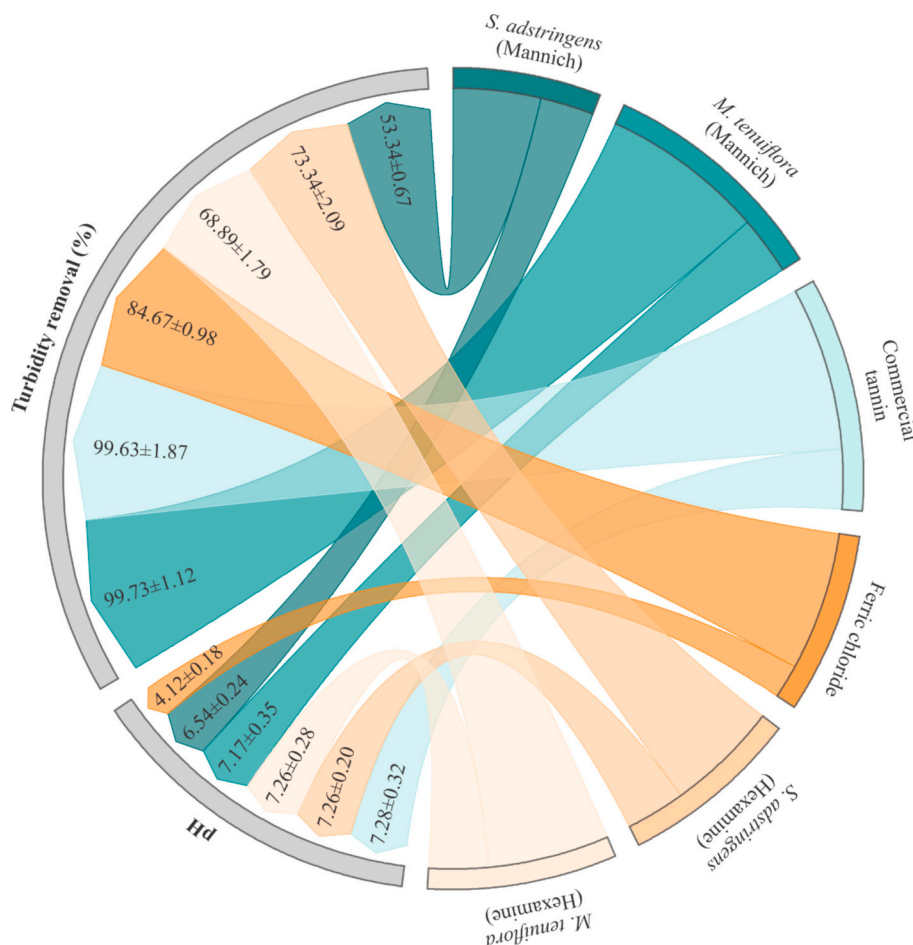


Fig. 3. Coagulant type, final pH, and turbidity removal efficiency (%), comparing *Mimosa tenuiflora* (Mannich and Hexamine), *S. adstringens* (Mannich and Hexamine), commercial tannin, and ferric chloride, using 50 mg of coagulant, 3 min of rapid mixing, and 30 min of slow mixing.

while the other coagulants tested maintained pH levels that were either comparable to or exceeded the initial measurement. This finding is advantageous for tannin-based coagulants, as it negates the necessity for alkalinizing agents to modify the pH, thus minimizing the production of residual sludge and lowering operational expenses [3,46]. Additionally, natural coagulants produce a reduced volume of biodegradable sludge and demonstrate performance that is less sensitive to fluctuations in water pH [47].

Fig. 3 illustrates that both cationization methods, whether employing the Mannich reaction or hexamine, produced comparable results for the two species examined. In terms of pH, the use of formaldehyde as a substitute in the cationization process yielded favorable results, demonstrating performance akin to that of commercial black wattle tannins, which are commonly utilized as coagulants in water treatment. Leiviskä and Santos [2] highlight that a significant benefit of tannin-based coagulants is their limited effect on altering water pH. Additionally, Fig. 3 illustrates that *Mimosa tenuiflora*, modified via the Mannich reaction, attained a turbidity removal rate exceeding 99%. This performance is on par with that of commercial black wattle tannins (99%) and significantly outperforms ferric chloride (84%). According to Alazaiza et al. [46], the required dosage of natural coagulants is lower than that of chemical ones. For this reason, ferric chloride exhibited higher turbidity removal when applied at a higher concentration (Fig. 4). The tannin-based coagulants exhibited enhanced efficacy compared to traditional coagulants, while also presenting benefits such as renewable sourcing, biodegradability, non-toxicity, non-corrosiveness, and cost efficiency [46,48].

The remarkable efficacy of *Mimosa tenuiflora* in reducing turbidity,

achieved with a relatively low concentration of coagulant, can likely be attributed to the high molecular weight of its tannins [49]. Observations during the coagulation process indicated that the tannins from *Mimosa* reacted swiftly, resulting in the formation of larger flocs in comparison to those produced by *Stryphnodendron adstringens*. According to Kweiner Tetteh and Rathilal [50], coagulants with higher molecular weight are more effective in agglomeration and floc formation. The outcomes for this species were both satisfactory and aligned with findings from Bello and Leiviskä [14], who reported that tannin-based coagulants maintained low residual turbidity even at reduced coagulant dosages of 25 and 50 mg L⁻¹.

For the concentration and agitation speeds used, it was possible to see that *S. adstringens*, through the Mannich reaction, showed a turbidity removal of 73%. However, this removal percentage can be explained by the low concentration of tannins used, since when 150 mg of the coagulant was used, the removal of water turbidity rose to 96%, as shown in Fig. 4. The need for high concentrations of tannin-based coagulants was also found by Beltrán-Heredia et al. [51] for *Acacia mearnsii*, where 150 mg of cationized tannins based on the Mannich reaction was required to achieve better turbidity removal.

Mimosa tenuiflora and *Stryphnodendron adstringens*, cationized with hexamine, achieved turbidity removal rates of 73% and 70%, respectively. The tannin-based coagulants produced in this study using hexamine as the cationizing agent did not form highly crosslinked polymeric structures, as formaldehyde, essential for promoting higher degrees of crosslinking, was not used. More extensively polymerized structures enable more effective interactions with suspended colloidal particles, facilitating the formation of larger and denser flocs [1].

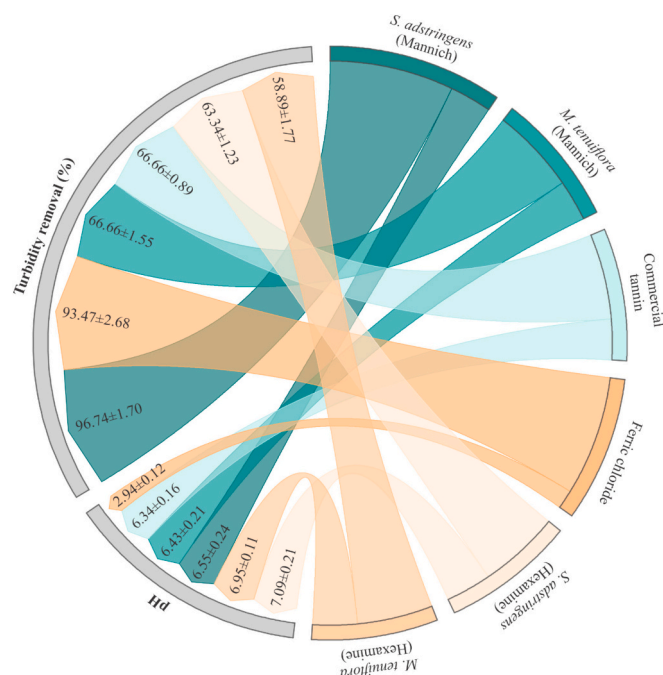


Fig. 4. Coagulant type, final pH, and turbidity removal efficiency (%), comparing *Mimosa tenuiflora* (Mannich and Hexamine), *S. adstringens* (Mannich and Hexamine), commercial tannin, and ferric chloride, using 150 mg of coagulant, 1 min of rapid mixing, and 10 min of slow mixing.

Similarly, Bello and Leiviskä [14], when using melamine as the nitrogen source in the Mannich reaction, observed that although the reaction occurred, the final product exhibited a low molecular weight. Therefore, it is likely that the coagulant derived from the hexamine-based reaction also resulted in a lower molecular weight polymer, which may explain its lower turbidity removal efficiency compared to coagulants produced via the traditional Mannich reaction.

Fig. 4 shows the final pH of the water after coagulation-flocculation tests conducted with different coagulants at a concentration of 150 mg, under rapid mixing at 100 rpm for 1 min followed by slow mixing at 30 rpm for 10 min. The coagulation-flocculation process is influenced by environmental factors such as pH, coagulant dosage, and both the duration and speed of mixing. Mixing time can affect the reaction rate and completeness, while shear forces acting on suspended particles also play a critical role in the formation and stability of flocs [52,53]. It can be observed that for the pH response variable, the pattern remained the same for the tannin-based coagulants, even with changes in concentration and agitation speeds. However, for ferric chloride, an increase in concentration led to a reduction in water pH, a pattern also observed by [54], who reported an inverse relationship between pH and ferric chloride dosage. With increasing coagulant concentration, the tannins from *Stryphnodendron adstringens* and *Mimosa tenuiflora*, cationized using the conventional method, achieved turbidity removal rates above 96%, outperforming ferric chloride, which reached 93%. Ferric chloride also showed improved efficiency at higher dosages. Similarly, in a study by [32], the use of tannin-based coagulants from *Terminalia catappa* L. resulted in optimal turbidity removal (96%) when applied at a higher concentration of 200 mg.

As shown by the results of this study, in addition to coagulant concentration, mixing time and speed have a significant influence on turbidity removal. Therefore, optimizing multiple operational parameters is essential to achieve maximum coagulation-flocculation efficiency. Karam et al. [55] tested different FeCl_3 dosages (0.25 – 1.5 g L^{-1}) across a range of pH values (1 – 11), with rapid mixing (50 – 250 rpm for 1 – 5 min) and slow mixing (10 – 50 rpm for 10 – 30 min), identifying optimal conditions at pH 9, with rapid mixing at 150 rpm for 1 min and slow mixing

at 30 rpm for 20 min . Similarly, Silva et al. [32] evaluated different doses of tannin-based coagulants from *Terminalia catappa* (50 – 200 mg L^{-1}), slow mixing durations (10 – 20 min), and sedimentation times (10 – 60 min), achieving optimal performance with 200 mg L^{-1} , 20 min of slow mixing at 30 rpm , and 30 min of sedimentation. These findings suggest that mixing time is a critical factor in optimizing water treatment performance, and that laboratory-scale testing is essential for determining ideal operational conditions for each coagulant [52].

Moreover, the chemical modification of tannins is a crucial factor to consider in the efficiency of the coagulation process. Machado et al. [1], in producing a formaldehyde-free tannin-based coagulant using ammonium hydroxide as a substitute, observed that although cationization occurred, the resulting coagulants exhibited lower molecular weight, which negatively affected turbidity removal. Similarly, in the present study, coagulants modified with hexamine also did not achieve satisfactory turbidity removal, despite FTIR analysis confirming that cationization had taken place. This reduced performance can be attributed to the fact that hexamethylenetetramine (hexamine) releases formaldehyde slowly and only under specific conditions, such as acidic environments or heating [56]. Thus, the absence of free formaldehyde in the reaction medium may lead to the formation of polymers with a lower degree of crosslinking, reduced molecular weight, and lower density of cationic charges, which compromises the coagulant's ability to neutralize suspended colloidal particles, ultimately reducing its coagulation efficiency.

However, it is expected that cationization using formaldehyde will show more effective results in destabilizing the colloidal particles of the water, since this compound allows the polymerization of the tannin together with a nitrogen compound, causing a large polymer structure to be formed, with positively charged nitrogen groups [38,57].

3.4. Practical applications and future research

The use of naturally derived products, when responsibly sourced, represents a powerful tool for promoting a more sustainable future, particularly when applied to environmental remediation or the improvement of industrial processes [58]. In the present study, the use of natural products for water treatment opens new opportunities for eco-friendly alternatives. Beyond creating new income streams—considering the significant impact of forest-derived products on the industrial sector—the use of natural products reduces reliance on synthetic chemicals and contributes positively to the water treatment sector [14].

The findings of this study can support decision-makers in both industrial and governmental sectors in opting for cost-effective and efficient alternatives. This aligns with the United Nations' Sustainable Development Goals (SDGs), particularly Goals 9 (Industry, Innovation, and Infrastructure) and 11 (Sustainable Cities and Communities), as well as, more specifically, Goals 6 (Clean Water and Sanitation) and 14 (Life Below Water) [59]. Given the increasing concerns regarding access to quality water and the potential challenges associated with it in the coming years, the pursuit of efficient and scalable alternatives is crucial to ensuring the effectiveness of water treatment strategies.

Future studies should focus on identifying tannins from other forest species across different regions that may serve the same purpose. This approach could lead to cost reductions and the development of natural water treatment products tailored to various parts of the world, particularly those most affected by water quality issues. Additionally, further investigations into different concentrations and treatment conditions should be conducted to enhance the efficiency of the tested products. In addition, studies that perform a more detailed characterization of these coagulants are also important for the field. These efforts would contribute to improving the economic and technical viability of natural coagulants for water treatment.

4. Conclusion

The results obtained from the coagulation-flocculation process indicated that both chemical modifications of the tannins from *Stryphnodendron adstringens* and *Mimosa tenuiflora* were successful and effective in removing water turbidity. However, the modification via the Mannich reaction yielded better result and favored the formation of larger and denser flocs.

Both *Stryphnodendron adstringens* and *Mimosa tenuiflora* exhibited turbidity removal efficiencies superior to the conventionally used coagulant, with results comparable to those of commercial tannin-based coagulants. Notably, the coagulant derived from *Mimosa tenuiflora* demonstrated greater efficiency when applied at lower concentrations and with longer mixing times, which is advantageous as it contributes to reducing operational costs in water treatment for public supply.

Although the cationization of these species using hexamine showed some effectiveness in turbidity removal, the results were inferior to those obtained through the Mannich reaction. This highlights the need for further research into this alternative route to optimize reaction conditions and improve the performance of the resulting coagulants.

CRediT authorship contribution statement

Bruna Rafaella Ferreira da Silva: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **João Gilberto Meza Ucella-Filho:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation. **Elias Costa de Souza:** Writing – review & editing, Writing – original draft. **Uasmim Lira Zidanes:** Writing – review & editing, Writing – original draft. **Renata Martins Braga:** Writing – review & editing, Writing – original draft, Methodology. **Tatiane Kelly Barbosa de Azevêdo:** Writing – review & editing, Writing – original draft, Methodology. **Fábio Akira Mori:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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